

# Lab 2: Analog Input and PWM

## Potentiometer, Serial Monitor, and LED Brightness

ENGR 1105 | Fall 2026

| Name | Partner |
|------|---------|
| Date | Section |

### Learning Objectives

By the end of this lab, you should be able to:

1. wire a potentiometer as a variable voltage divider;
2. read an analog input using `analogRead()`;
3. inspect raw measurements using the Serial Monitor;
4. estimate input voltage from an analog reading;
5. use `map()` to convert one numerical range to another;
6. control LED brightness with PWM and `analogWrite()`; and
7. create threshold-based output behavior.

### 1. Lab Organization

Lab 2 is completed in two class meetings:

| Part    | Main work                                                                                       |
|---------|-------------------------------------------------------------------------------------------------|
| Lab 2.1 | Wire a potentiometer, read analog values, use the Serial Monitor, and calculate voltage         |
| Lab 2.2 | Build a PWM LED circuit, control brightness, create thresholds, and complete a design challenge |

## 2. Materials

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| Item                      | Purpose                                  |
|---------------------------|------------------------------------------|
| Arduino Uno               | Controller                               |
| USB data cable            | Power and program upload                 |
| Solderless breadboard     | Temporary circuit construction           |
| Jumper wires              | Electrical connections                   |
| Potentiometer             | Variable analog input                    |
| LED                       | Visual output                            |
| 330 $\Omega$ resistor     | Limits LED current                       |
| Computer with Arduino IDE | Code editing, upload, and Serial Monitor |

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### Before Applying Power

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1. Disconnect the USB cable before making major wiring changes.
2. Never connect 5V directly to GND.
3. Verify the three potentiometer connections before applying power.
4. Use a resistor in series with the LED.
5. Remove power immediately if the Arduino becomes hot or repeatedly disconnects.

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# Lab 2.1

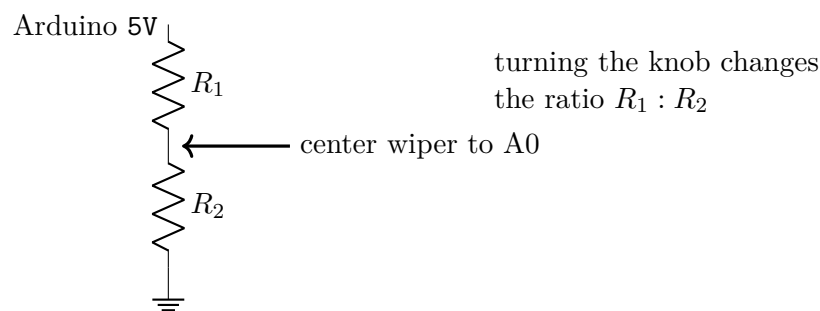
## Potentiometer and Serial Data

Reading a Variable Analog Input

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### 3. Part A: Understand the Potentiometer

A potentiometer is a three-terminal variable resistor. The two outside terminals connect across a voltage source. The center terminal, called the **wiper**, provides an adjustable voltage.



### Wiring Steps

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1. Disconnect the Arduino from USB power.
2. Place the potentiometer on the breadboard.
3. Connect one outside potentiometer terminal to Arduino 5V.
4. Connect the other outside terminal to Arduino GND.
5. Connect the center terminal to analog pin A0.
6. Inspect the wiring before reconnecting USB power.

### Potentiometer Wiring

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The two outside terminals may be reversed without damaging the potentiometer; reversing them changes which turning direction increases the reading. Do not connect the center terminal directly between 5V and GND.

### 4. Part B: Read the Raw Analog Value

```
1 const int potPin = A0;
2
3 void setup() {
```

```
4   Serial.begin(9600);  
5 }  
6  
7 void loop() {  
8   int sensorValue = analogRead(potPin);  
9   Serial.println(sensorValue);  
10  delay(100);  
11 }
```

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## Instructions

1. Enter the program in a new Arduino sketch.
2. Verify and upload the program.
3. Open Tools > Serial Monitor.
4. Set the Serial Monitor baud rate to 9600.
5. Turn the potentiometer slowly from one end to the other.
6. Observe the smallest, middle, and largest readings.

## Raw Analog Measurements

Record representative values. Exact readings may vary slightly.

| Knob position      | Analog reading | Expected range |
|--------------------|----------------|----------------|
| Near one end       |                | close to 0     |
| Near the middle    |                | close to 512   |
| Near the other end |                | close to 1023  |

Does the reading increase when you turn clockwise or counterclockwise?

Does the reading remain perfectly constant when the knob is not moving?

## Checkpoint 1: Analog Reading

Demonstrate that turning the potentiometer changes the Serial Monitor reading across most of the range from 0 to 1023. Explain which wire carries the adjustable voltage to A0.

## 5. Part C: Convert the Reading to Voltage

For a nominal 5 V reference, estimate the input voltage using

$$V_{\text{in}} = \frac{\text{sensorValue}}{1023} (5 \text{ V}).$$

Modify the program:

```

1  const int potPin = A0;
2
3  void setup() {
4    Serial.begin(9600);
5  }
6
7  void loop() {
8    int sensorValue = analogRead(potPin);
9    float voltage = sensorValue * (5.0 / 1023.0);
10

```

```
11 Serial.print("Reading: ");
12 Serial.print(sensorValue);
13 Serial.print("    Voltage: ");
14 Serial.println(voltage);
15
16 delay(100);
17 }
```

### Reading and Estimated Voltage

Complete the measured-value columns, then calculate the expected voltage using the formula.

| Position | Reading | Printed voltage | Your calculation |
|----------|---------|-----------------|------------------|
| Low      |         |                 |                  |
| Middle   |         |                 |                  |
| High     |         |                 |                  |

Show one voltage calculation:

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Why does the program use 5.0 and 1023.0 instead of only whole numbers?

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### Checkpoint 2: Reading and Voltage

Show the raw reading and estimated voltage in the Serial Monitor. Explain why a midrange reading corresponds to approximately 2.5 V.

## 6. Part D: Observe Measurement Variation

Hold the potentiometer still and watch the Serial Monitor for approximately ten seconds.

### Measurement Variation

What is the smallest value observed? \_\_\_\_\_

What is the largest value observed? \_\_\_\_\_

Variation:

largest reading – smallest reading = \_\_\_\_\_

List two possible reasons the value may vary slightly.

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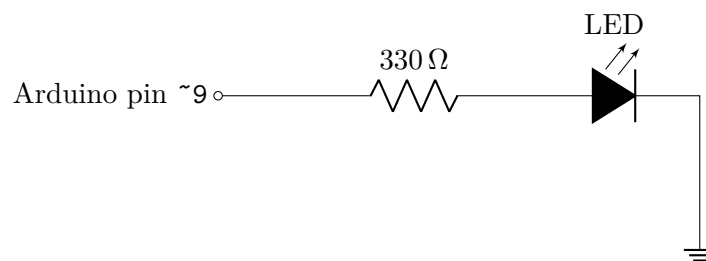
# Lab 2.2

## PWM LED Brightness and Thresholds

Using Analog Input to Control an Output

### 7. Part E: Add the PWM LED Circuit

Keep the potentiometer connected to A0. Add an external LED controlled by PWM pin 9.



### Wiring Steps

1. Disconnect USB power.
2. Place the LED on the breadboard with its leads in separate terminal groups.
3. Connect PWM pin ~9 to one end of the 330 Ω resistor.
4. Connect the other resistor end to the LED's long lead.
5. Connect the LED's short lead to Arduino GND.
6. Keep the potentiometer connected to 5V, GND, and A0.
7. Inspect the circuit before reconnecting USB power.

### PWM Pin and LED Safety

Use a PWM-capable pin marked with a tilde. The LED still requires a series resistor even though PWM switches the output rapidly.

### 8. Part F: Control LED Brightness

```
1 const int potPin = A0;
2 const int ledPin = 9;
3
4 void setup() {
5   pinMode(ledPin, OUTPUT);
```



```
6   Serial.begin(9600);
7 }
8
9 void loop() {
10   int sensorValue = analogRead(potPin);
11   int brightness = map(sensorValue, 0, 1023, 0, 255);
12
13   analogWrite(ledPin, brightness);
14
15   Serial.print("Input: ");
16   Serial.print(sensorValue);
17   Serial.print("    PWM: ");
18   Serial.println(brightness);
19
20   delay(20);
21 }
```

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**Instructions**

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1. Upload the brightness-control program.
2. Turn the potentiometer slowly through its full range.
3. Observe the LED brightness.
4. Observe the input and PWM values in the Serial Monitor.
5. Stop at low, middle, and high positions and record the values.

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**Brightness Measurements**

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| Knob position | Input value | PWM value | LED appearance |
|---------------|-------------|-----------|----------------|
| Low           |             |           |                |
| Middle        |             |           |                |
| High          |             |           |                |

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What does `map(sensorValue, 0, 1023, 0, 255)` do?

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Why can the LED appear partly bright even though pin 9 is a digital output?

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**Checkpoint 3: Variable Brightness**

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Demonstrate smooth LED brightness control. Explain the relationship among the analog reading, the mapped PWM value, and the apparent brightness.

**9. Part G: Create Threshold-Based Behavior**

Replace the smooth brightness control with three distinct output levels.

```
1  const int potPin = A0;
2  const int ledPin = 9;
3
4  void setup() {
5      pinMode(ledPin, OUTPUT);
6  }
7
8  void loop() {
9      int sensorValue = analogRead(potPin);
10
11     if (sensorValue < 300) {
12         analogWrite(ledPin, 0);
13     } else if (sensorValue < 700) {
14         analogWrite(ledPin, 128);
15     } else {
16         analogWrite(ledPin, 255);
17     }
18 }
```

## Threshold Results

| Input range     | PWM command | Expected LED state |
|-----------------|-------------|--------------------|
| Below 300       | 0           |                    |
| 300 through 699 | 128         |                    |
| 700 and above   | 255         |                    |

At what approximate knob positions does the LED change state?

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Why is `else if` useful when the program has more than two operating regions?

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## Checkpoint 4: Threshold Control

Demonstrate the OFF, medium-brightness, and full-brightness regions. Explain how the program chooses one region.

## 10. Troubleshooting

| Problem                                          | Check first                                                                         |
|--------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Serial Monitor shows nothing</b>              | Confirm <code>Serial.begin(9600)</code> , correct port, and matching baud rate      |
| <b>Analog reading does not change</b>            | Check potentiometer outside terminals, center wiper, A0, 5V, and GND                |
| <b>Reading changes in the opposite direction</b> | Reverse the two outside potentiometer connections, or accept the reversed direction |
| <b>LED does not light</b>                        | Check LED polarity, resistor, pin 9, ground, and <code>pinMode()</code>             |
| <b>LED is only ON or OFF</b>                     | Confirm use of PWM pin ~9 and <code>analogWrite()</code>                            |
| <b>Arduino becomes hot</b>                       | Disconnect immediately and inspect for a short circuit                              |

## 11. Part H: Design Challenge

### Design Challenge

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Choose one challenge:

1. **Reverse brightness:** Make the LED become dimmer as the analog reading increases.
2. **Adjustable blink rate:** Use the potentiometer to control the delay between LED changes.
3. **Useful operating range:** Keep the LED OFF below an input of 200, then map the range 200–1023 to PWM values 0–255.

### Design Record

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Which challenge did you choose?

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Describe the program changes.

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Record one test input and the resulting output.

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What problem did you encounter, and how did you troubleshoot it?

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## 12. Final Reflection

Explain how Lab 2 demonstrates the sense-decide-act model. Identify the sensed quantity, the program decision or calculation, and the physical action.

### 13. Submission Checklist

- ☐ Completed Lab 2 instruction sheet
- ☐ Raw analog and voltage measurements
- ☐ Brightness and threshold observations
- ☐ Clear photo of the final circuit
- ☐ Final Arduino code
- ☐ Completed design challenge and reflection